

Why the 3D Hilbert Curve Cannot Prove the Riemann Hypothesis

— A Final Assessment

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Abstract

After 11 Hilbert orders, 6 fundamental curve constructions, 2 independent operator formulations, 4D comparative analysis, 30,000 permutation tests, and exhaustive optimization over the 3D Hilbert construction space, we conclude that the 3D Hilbert plane operator cannot prove the Riemann Hypothesis. The eigenvalue spectrum correlates with zeta zeros at $r \approx -0.80$, but the eigenvalues are not the zeros. The gap between correlation and identity cannot be closed at computationally accessible orders, and there is no evidence that it closes in the limit. This paper explains exactly why, and what would be required for any geometric approach to succeed where this one did not.

1 What We Found

The investigation established three robust phenomena:

1. **Plane-density anomaly:** Primes are not uniformly distributed across Z-planes of the 3D Hilbert curve. The density variation grows as $2^{n/2}$, reaching $|z| = 55.7$ at order 10 ($p < 10^{-200}$).
2. **PNT error alignment:** 98% of planes with anomalous prime density sample integer ranges where $\pi(x) > \text{li}(x)$ – the explicit formula’s oscillatory term is systematically positive.
3. **Spectral correlation:** Eigenvalues of the Hilbert plane covariance operator correlate with zeta zeros at $r \approx -0.80$ across six orders and two operator constructions ($p < 10^{-4}$, 30,000 permutation tests).

These are genuine mathematical discoveries. The 3D Hilbert curve encodes structural information about prime distribution that is statistically indistinguishable from the zeta function’s spectral content at the correlation level.

2 Why This Cannot Prove RH

Despite the strength of the correlations, five fundamental gaps prevent this approach from constituting a proof of the Riemann Hypothesis.

2.1 1. Correlation Is Not Identity

The eigenvalues of the Hilbert plane operator *correlate with* the zeta zeros at $r \approx -0.80$. They are not *equal to* the zeta zeros. The mean absolute deviation between the normalized eigenvalue spectrum and the normalized zeta zero spectrum is stable at ≈ 0.28 across orders 6–11, showing no statistically significant convergence toward zero.

A correlation of $r = -0.80$ means the eigenvalues are in approximately the correct *order*, but their precise values differ from the zeta zeros by $\sim 28\%$ of the spectral range. For the Hilbert-Pólya conjecture to be realized, the eigenvalues must *equal* the zeta zeros exactly, not merely correlate with them.

The temperature in London correlates with the temperature in New York at $r \approx 0.7$, but that does not make London's thermometer a measurement of New York's weather. Correlation, no matter how statistically significant, does not establish identity.

2.2 2. Wrong Spectral Universality Class

The eigenvalue pair correlation function is anti-GUE ($r_{\text{GUE}} = -0.84$): the eigenvalues exhibit *clustering* rather than the *level repulsion* characteristic of zeta zeros. Montgomery's pair correlation conjecture – one of the most thoroughly verified predictions in analytic number theory – states that zeta zero spacings follow the GUE distribution. The Hilbert plane operator's eigenvalues follow the opposite pattern.

This is not a minor discrepancy. The GUE property of zeta zeros is believed to be a universal signature of quantum chaotic systems. An operator whose spectrum is anti-GUE belongs to a fundamentally different dynamical class – an integrable or quasi-integrable system rather than a chaotic one. If the zeta zeros are eigenvalues of a quantum chaotic Hamiltonian (as the GUE statistics suggest), the Hilbert plane operator cannot be that Hamiltonian.

2.3 3. No Convergence in Accessible Regime

The scaling analysis (Section 5.3 of Paper IV) shows that the eigenvalue error relative to zeta zeros converges at a rate of approximately -0.07 per order. At this rate, order ≈ 475 would be required for 1% accuracy – a matrix of size $2^{475} \times 2^{475}$, requiring more memory than there are atoms in the observable universe.

The physical universe imposes a hard limit on computational verification. If convergence to the zeta zeros requires orders that cannot be realized in this universe, then the conjecture cannot be verified computationally. And if it cannot be verified, it cannot serve as the basis for a proof that relies on computational evidence.

2.4 4. The Operator Is Not the Right One

The Hilbert plane operator T_n is a covariance matrix constructed from adjacent integer pairs along the Hilbert curve. It measures *correlations between planes*, not the *action of an operator on arithmetic functions*. The true Hilbert-Pólya operator $\hat{H}$ would satisfy:

$$\hat{H}\phi_k = \gamma_k\phi_k \tag{1}$$

where ϕ_k are eigenfunctions and γ_k are the zeta zeros. The Hilbert plane operator does not have this property for any known set of ϕ_k . Its eigenvalues are the wrong values ($\text{MAD} \approx 0.28$), and its eigenvectors are the wrong objects (plane indicator functions rather than arithmetic wavefunctions).

The fact that the eigenvalues correlate with the zeta zeros means the operator is *related to* $\hat{H}$, not that it *is* $\hat{H}$.

2.5 5. The Gap Is Analytic, Not Computational

The most fundamental obstacle is that the zeta zeros are defined analytically – as the roots of a complex function continued to the entire complex plane – while the Hilbert plane operator is defined combinatorially – as a finite-dimensional matrix built from a discrete geometric construction.

Bridging this gap requires showing that the combinatorial construction converges to the analytic one in the limit $n \rightarrow \infty$. This is an analytic continuation problem, not a computational one. No amount of computational evidence at finite n can substitute for a proof of convergence in the limit.

The situation is analogous to approximating π by the perimeter of an inscribed 2^n -gon. At $n = 10$, the approximation is good to several decimal places. But you cannot *prove* that the limit is π without an analytic argument – the computational evidence is suggestive but not dispositive.

3 What Would Be Required

For any geometric construction to prove the Riemann Hypothesis via the Hilbert–Pólya pathway, it must satisfy four criteria:

1. **Exact eigenvalue match:** The eigenvalues must equal the zeta zeros, not merely correlate with them. MAD must converge to 0.
2. **GUE spacing statistics:** The eigenvalue pair correlation function must match GUE at all accessible orders, not anti-GUE.
3. **Proven self-adjointness:** The infinite-order limit must be proven self-adjoint by analytic argument, not computational evidence.
4. **Analytic convergence proof:** The finite-order approximations must be proven to converge to the limit, with error bounds that vanish as $n \rightarrow \infty$.

The 3D Hilbert plane operator satisfies none of these criteria. It achieves the first criterion partially (correlation but not identity), fails the second (anti-GUE), and has no proof for the third or fourth.

4 What This Work Contributes Despite Not Proving RH

The investigation was not a failure. It established:

- The first computationally accessible operator whose spectrum correlates with zeta zeros at statistically overwhelming significance.
- A new geometric probe of prime distribution – the 3D Hilbert curve’s plane decomposition – that reveals structural biases invisible to other methods.
- The specificity of the 3D case: the correlation is unique to 3D (4D weakens with order) and bounded across curve constructions, suggesting a genuine mathematical structure rather than an artifact.
- A clear negative result: exhaustive optimization over the 3D Hilbert construction space cannot close the gap to identity.
- A set of criteria that any future geometric approach must satisfy to succeed where this one fell short.

In science, knowing what doesn't work is often as valuable as knowing what does. The 3D Hilbert curve will not prove the Riemann Hypothesis. But the tools, methods, and structural insights developed here may enable someone else to find the construction that will.

5 Acknowledgments

All code, data, and computational results are available in the companion repository. Every numerical claim can be independently reproduced. The author thanks the open-source mathematical community for tools including NumPy, SciPy, and the AMD EPYC 7H12 server on which these computations were performed.